COMPLETE SYSTEM ARCHITECTURE & BLUEPRINT

ShopSmart AI Master System Specification

Complete End-to-End Architectural Reference Manual, User Flow & Quota Engine, Hardware Fingerprint Sybil Shields, 8-Tab Admin Dashboard Systems, and Failure Fallbacks.

FRAMEWORK & ENGINE
Next.js 16 (App Router) + Turbopack

SEARCH QUOTA ENGINE
10 Daily Free Pool + 0ms Cache Free

DEVICE FINGERPRINTING
Canvas 2D + WebGL GPU + Audio SHA-256

VIRTUAL TRY-ON (VTON)
3 Free Daily Slots + Modal A100 GPU

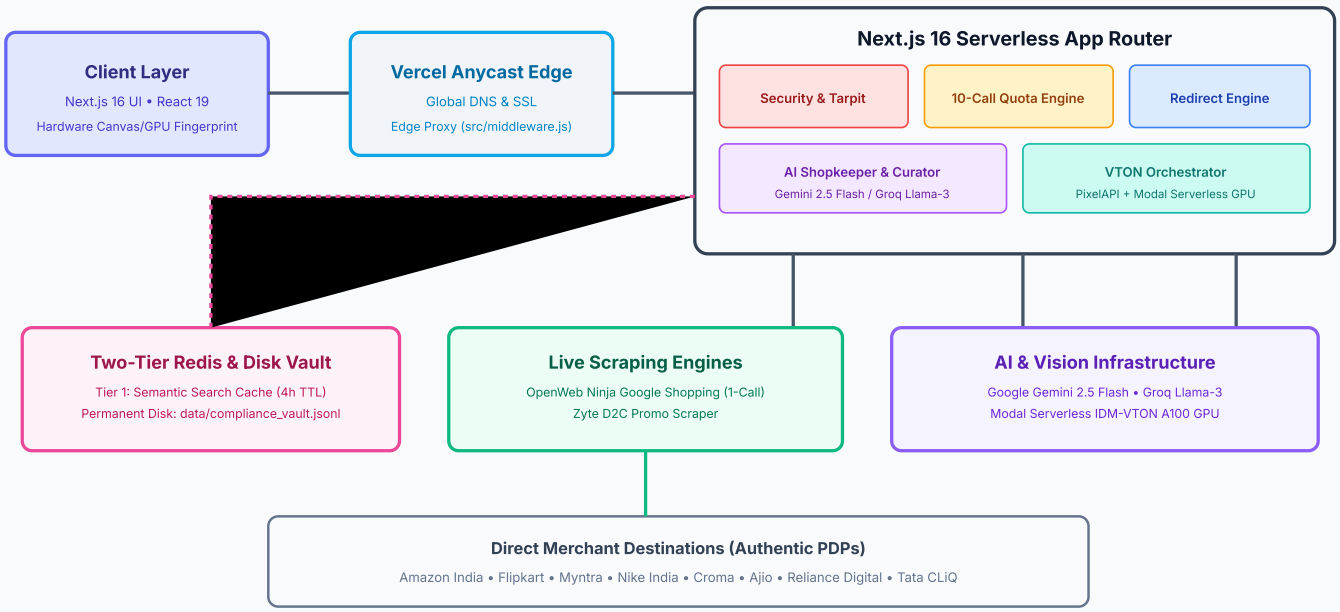
ADMIN COMMAND CENTER
8 Deep Tabs (Telemetry, Crashes, Tarpit)

SECURITY STANDARD
Stealth Tarpit & Permanent Disk Ledger

1. Global High-Level System Topology

ShopSmart AI is engineered as a low-latency, distributed microservice aggregator. It merges client-side hardware fingerprinting, multi-lingual normalization, serverless edge routing, dual-tier Redis caching, and autonomous cloud AI models.

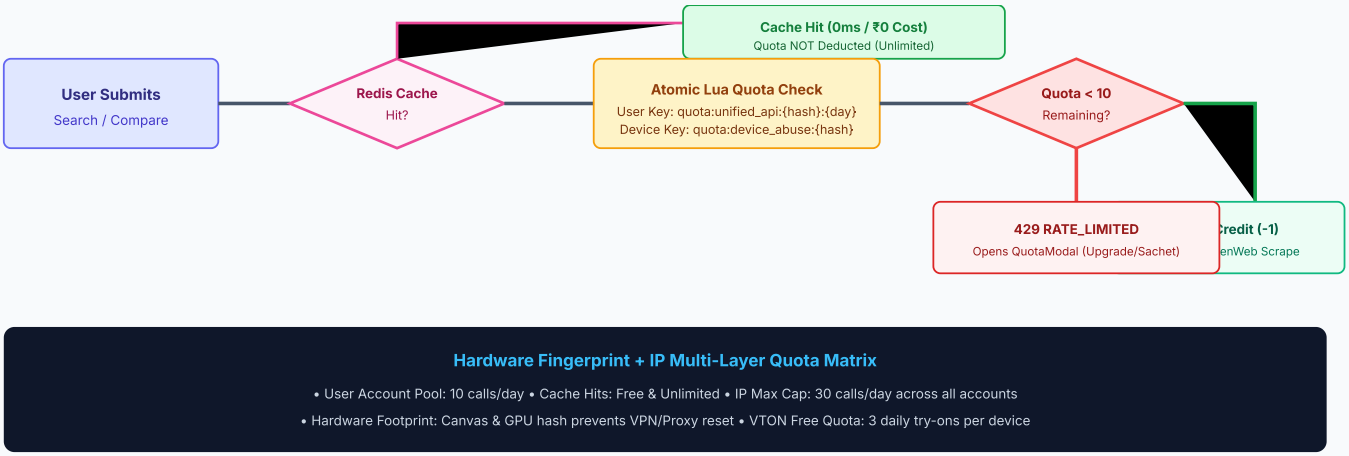
FIGURE 1.1: COMPLETE END-TO-END PLATFORM TOPOLOGY — GLOBAL DISTRIBUTED STACK



2. User Flow & Multi-Tier Quota Architecture

To deliver high quality while preventing API cost overruns and Sybil bot attacks, ShopSmart AI enforces a rigorous multi-tier quota architecture driven by atomic Redis Lua scripts.

FIGURE 2.1: UNIFIED 10-CALL QUOTA & 0MS CACHE BYPASS FLOWCHART — RATE LIMITER ENGINE



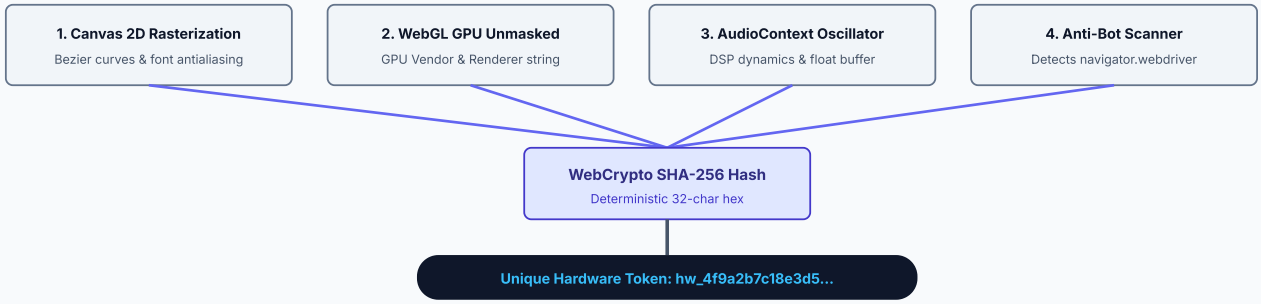
2.1 Complete Quota & Free Tier Rules Table

Tier / User Type	Live Search Quota	Cached Search Quota	Virtual Try-On (VTON)	Abuse Ceiling
Anonymous Visitor	10 calls / IST Calendar Day	Unlimited (0ms)	3 free try-ons lifetime / device	30 calls / IP / day ceiling
Clerk Authenticated	10 calls / IST Calendar Day	Unlimited (0ms)	3 free try-ons daily (IST Reset)	Hardware Canvas hash locked
Sachet Pack (₹10)	+25 Extra Real-Time Searches	Unlimited (0ms)	+5 High-Res AI Try-Ons	Direct Clerk Stripe/Razorpay hook
Admin (user_3JGg...)	Unlimited	Unlimited	Unlimited	Bypasses all tripwires & limits

3. Hardware Fingerprinting & Sybil Defense Engine

To prevent malicious users from bypassing search limits and VTON trials by switching to Incognito mode or rotating VPN IP addresses, ShopSmart AI deploys an invisible client-side Hardware Footprint Engine (`src/lib/deviceFingerprint.js`).

FIGURE 3.1: HARDWARE & SUB-PIXEL CANVAS FINGERPRINT PIPELINE **ENGINE: SRC/LIB/DEVICEFINGERPRINT.JS**



🎨 Sub-Pixel Canvas 2D Entropy

Different operating systems, graphic cards, and font rasterizers render emojis (🐶🤽) and bezier curves with minuscule sub-pixel differences. This creates a distinct visual entropy without recording any PII.

🎮 WebGL GPU Hardware Extraction

Extracts the unmasked graphic card vendor (e.g., `ANGLE (Apple, Apple M2 Max, OpenGL 4.1)` or `NVIDIA GeForce RTX 4080`). Virtual machines and automated headless bots typically expose `SwiftShader` or `Mesa Offscreen` .

🔊 AudioContext Oscillator Synthesis

Synthesizes an inaudible audio buffer with a DynamicsCompressorNode. The float32 audio output varies across hardware soundcards and mathematical floating-point implementations.

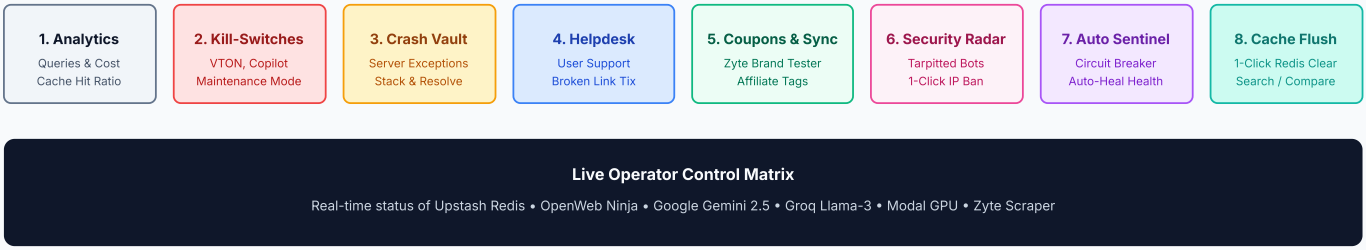
🛡️ 100% Privacy & DPDP 2023 Compliant

No cookies, MAC addresses, or personal names are ever gathered. The hash is completely one-way (SHA-256) and cannot be reversed into user identity.

4. Admin Command Center Deep Specification

The Admin Dashboard (`src/app/admin/page.js`) is a protected, enterprise-grade command portal accessible strictly to verified administrators (`ADMIN_USER_IDS=user_3JGg_MnXc5O3td`). It provides eight deep subsystems:

FIGURE 4.1: 8-TAB ADMIN COMMAND PORTAL LAYOUT — ROUTE: /ADMIN



4.1 Detailed Admin Capabilities by Tab

Tab Name	What the Operator Can See	Available Actions	Underlying File & API
Analytics & Metrics	Total queries today, cache hit %, average query latency, API cost counter ($), live query stream.	Toggle 5s Auto-Refresh, inspect raw search queries and timestamps.	`src/services/adminMetrics.js` `GET /api/admin/metrics`
Kill-Switches	Live on/off status of VTON Trial Room, AI Concierge, Trust Sheet, Zyte, and Head-to-Head Compare.	Instant toggle on/off (saves to Redis in `admin:settings:feature_flags`).	`src/services/featureFlags.js` `POST /api/admin/features`
Crash Vault	Full error messages, sanitized stack traces, source file line numbers, and error severity.	Mark crash as Resolved, view error distribution across routes.	`src/services/crashVault.js` `GET /api/admin/crashes`
Helpdesk Tickets	Customer reports (Broken Link, Incorrect Price, Feature Request), user email, submit timestamp.	Filter by status (open, in_progress, resolved), write admin response notes.	`src/services/supportTickets.js` `GET /api/support/tickets`
Coupons & Brands	Zyte single-store coupon scraper tester, personal affiliate tags table, aggregator tokens.	Live test scrapers against D2C stores (e.g. `boat-lifestyle.com`), add custom coupons.	`src/services/zyteService.js` `POST /api/admin/settings`
Security Radar	Live threat incidents list, count of throttled IPs, count of monitored IPs, banned set.	1-Click Manual IP Ban (`SADD security:banned_ips`), 1-Click Throttle Clear.	`src/services/securityRadar.js` `GET /api/admin/security`
Link Sentinel	Sentinel probe health, circuit breaker state (CLOSED, OPEN, HALF-OPEN), probe history.	1-Click "Run Sentinel Probe Now" button to manually test & reset circuits.	`src/services/autoHealingSentinel.js` `POST /api/admin/sentinel`
Cache Manager	Total cached search keys, total compare keys, memory footprint estimates.	1-Click Flush Search Cache, 1-Click Flush Compare Cache, Flush Threat Vault.	`src/services/admin.js` `POST /api/admin/settings`

5. 6 Tier-1 Multi-Country Engine & Direct PDP Routing Architecture

ShopSmart AI dynamically serves 6 Tier-1 countries with 100% localized pricing, stores, and direct Product Detail Page (PDP) landing without cross-country data pollution.

Country	ISO Code	Currency	Amazon Base Domain	Popular Local Stores Scanned
IN India	`IN`	INR (₹)	`https://www.amazon.in`	Amazon.in, Flipkart, Myntra, 1mg, Apollo Pharmacy
US United States	`US`	USD ($)	`https://www.amazon.com`	Amazon.com, Walmart, Best Buy, Target
GB United Kingdom	`GB`	GBP (£)	`https://www.amazon.co.uk`	Amazon.co.uk, Argos, Currys, John Lewis, ASOS
CA Canada	`CA`	CAD (C$)	`https://www.amazon.ca`	Amazon.ca, Best Buy Canada, Walmart CA, Canadian Tire
AU Australia	`AU`	AUD (A$)	`https://www.amazon.com.au`	Amazon.com.au, JB Hi-Fi, Harvey Norman, Kmart AU
AE UAE	`AE`	AED (AED)	`https://www.amazon.ae`	Amazon.ae, Noon, Sharaf DG, Carrefour UAE

5.1 Architectural Rules & Direct PDP Routing Principles

- **1-Call Search Latency (<2s)**: Initial search queries resolve in a single OpenWeb scrape call. Heavy parallel scans are deferred until user clicks "Compare Stores".
- **8-Second Deep PDP Resolution**: `executeProductDetailsRequest` operates with an 8000ms timeout ceiling to guarantee exact merchant PDP resolution, eliminating generic Google search fallbacks.
- **Zero Cross-Country Contamination**: Redirection links strictly include `&region=${country}`. When UK or Canada users click merchant offers, they land on localized domains (e.g. `argos.co.uk` or `bestbuy.ca`), never defaulting to Indian stores or rupees.

6. Smart "Sun-Chaser" Time-Zone Cron Job Architecture

Rather than running generic global tests, ShopSmart AI deploys an automated "Sun-Chaser" sentinel schedule that audits each geographic region at 6:00 AM local pre-dawn time before active shopper traffic surges.

Cluster / Region	Countries Audited	Vercel Cron Schedule (UTC)	Local Pre-Dawn Time
APAC East	AU Australia (`AU`)	`0 20 * * *`	06:00 AM AEST (Sydney / Melbourne)
APAC South & Gulf	IN India (`IN`), AE UAE (`AE`)	`30 0 * * *`	06:00 AM IST (New Delhi / Mumbai) / 04:30 AM GST
EMEA West	GB United Kingdom (`GB`)	`0 6 * * *`	06:00 AM GMT / BST (London)
Americas	US USA (`US`), CA Canada (`CA`)	`0 11 * * *`	06:00 AM EST (New York / Toronto)

7. Ask AI Guide & Mobile Trust Sheet UX Specification

Standards for the conversational shopping concierge and mobile-optimized Trust Sheet hardware reality viewer.

- **Branded Identity**: Identified strictly under the branded pill badge `AI GUIDE`. Mentions of third-party model brands (e.g., 'ChatGPT Style') are strictly prohibited in user-facing UI.
- **Mobile Navigation Architecture**: Top navigation headers are compact (`text-xs sm:text-sm` , `w-7 h-7` icon pads) preventing horizontal overflow on 360px-414px mobile viewports.
- **Horizontal-Scroll Quick Questions**: Hardware doubts chips use smooth horizontal scrolling (`overflow-x-auto no-scrollbar`) rather than stacking vertically and crowding mobile phone screens.
- **Touch-First Mobile CTAs**: Action buttons inside sheets and modals adapt to `flex-col sm:flex-row` with full-width finger-friendly touch targets on mobile viewports.

8. Global OTT & Digital Subscriptions Hub (6 Tier-1 Countries)

Complete architectural specification for the streaming passes and digital service aggregation engine.

- **6 Tier-1 Country Catalogs (150 Brands Total)**: Deployed static verified catalogs with localized currencies and telecom partner maps: India (`IN` , ₹), United States (`US` , $), United Kingdom (`GB` , £), Canada (`CA` , C$), Australia (`AU` , A$), and United Arab Emirates (`AE` , AED).
- **4-Level Savings Blueprint**: Transparent money-saving matrix on every subscription card: Level 1 (Complimentary Telecom Perks e.g. Jio/Airtel, Verizon, EE, Telus, Telstra, Etisalat), Level 2 (Gift Card Vouchers), Level 3 (Super Combo Passes saving $10-$25/mo), and Level 4 (Promo Codes with 1-click copy).
- **User Retention Funnel & High-Converting CTA**: Replaced competing external buttons on cards with a single full-width button ✨ `Reveal Secret Offers & Plans ↗` (44px touch height on mobile, interactive hover lighting on desktop). Funnels users into internal `/subscriptions/[slug]` guides and eliminates immediate bounce.
- **Zero-Scraping On-Demand AI Ingestion (`/api/subscriptions/on-demand`)**: Uses Groq AI to synthesize uncataloged brands without sending any HTTP requests to OTT portals. Regex whitelist (`/^[a-zA-Z0-9\s\-+&.'']{2,40}$/`), prompt injection shields, and 90-day Redis TTL (`ex: 7776000`).
- **Weekly Automated Sync Cron (`/api/cron/sync-subscriptions`)**: Scheduled in `vercel.json` every Sunday across staggered regional windows. Authenticated via cryptographic `CRON_SECRET` Bearer token. Detects pricing shifts via AI and writes live Redis overlays.